

BEYOND NODAL DAMPING

WHEN STABLE REPLACEMENTS FAIL TOGETHER

Minimal Incompatibility Hypergraphs for Safe SG → IBR Transition Planning

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1 MOTIVATION & QUESTION

Replacing SG units changes the full portfolio. A stable endpoint can still contain an unstable intermediate coalition.

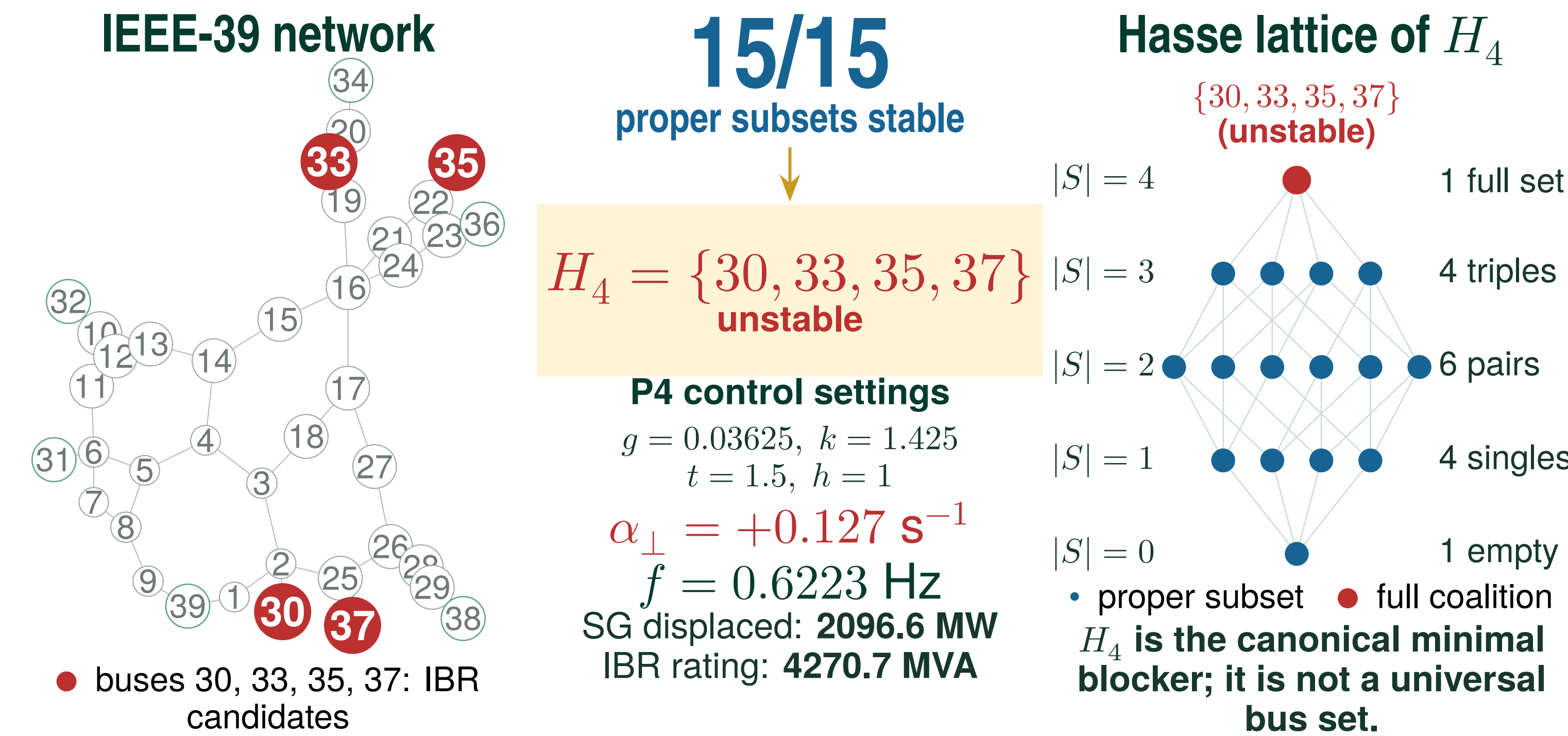


Which replacement combinations are incompatible, why do they fail, and how can paths avoid them?

$$\mathcal{H}(\theta) = \text{Min}_{\subseteq} \{S : \alpha_{\perp}(S, \theta) \geq 0\}$$

Edges mark minimal unstable coalitions.

2 FLAGSHIP RESULT: IEEE-39



3 STABLE TARGET $\neq$ SAFE PATH

Three nested guarantees for a target portfolio T :

A FINAL STATE STABLE $\alpha_{\perp}(T, \theta) < 0$	B SOME ORDER IS SAFE Stable prefix chain exists	C EVERY ORDER SAFE No blocker in T .
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$$C \Rightarrow B \Rightarrow A$$

Converse implications do not hold.

$$C(T) \Leftrightarrow \forall e \in \mathcal{H}(\theta), e \not\subseteq T$$

Exact $\mathcal{H}$ recovered on the frozen F10 set

Counts of 52 operating points

first-order modal 22/52

additive 31/52

pairwise 38/52

15/15
smaller portfolios stable
all strict subsets

$H_4 = \{30, 33, 35, 37\}$
canonical witness
not the universal blocker

$+0.127 \text{ s}^{-1}$
critical growth rate
canonical portfolio

0.6223 Hz
critical frequency
nominal IEEE-39 case

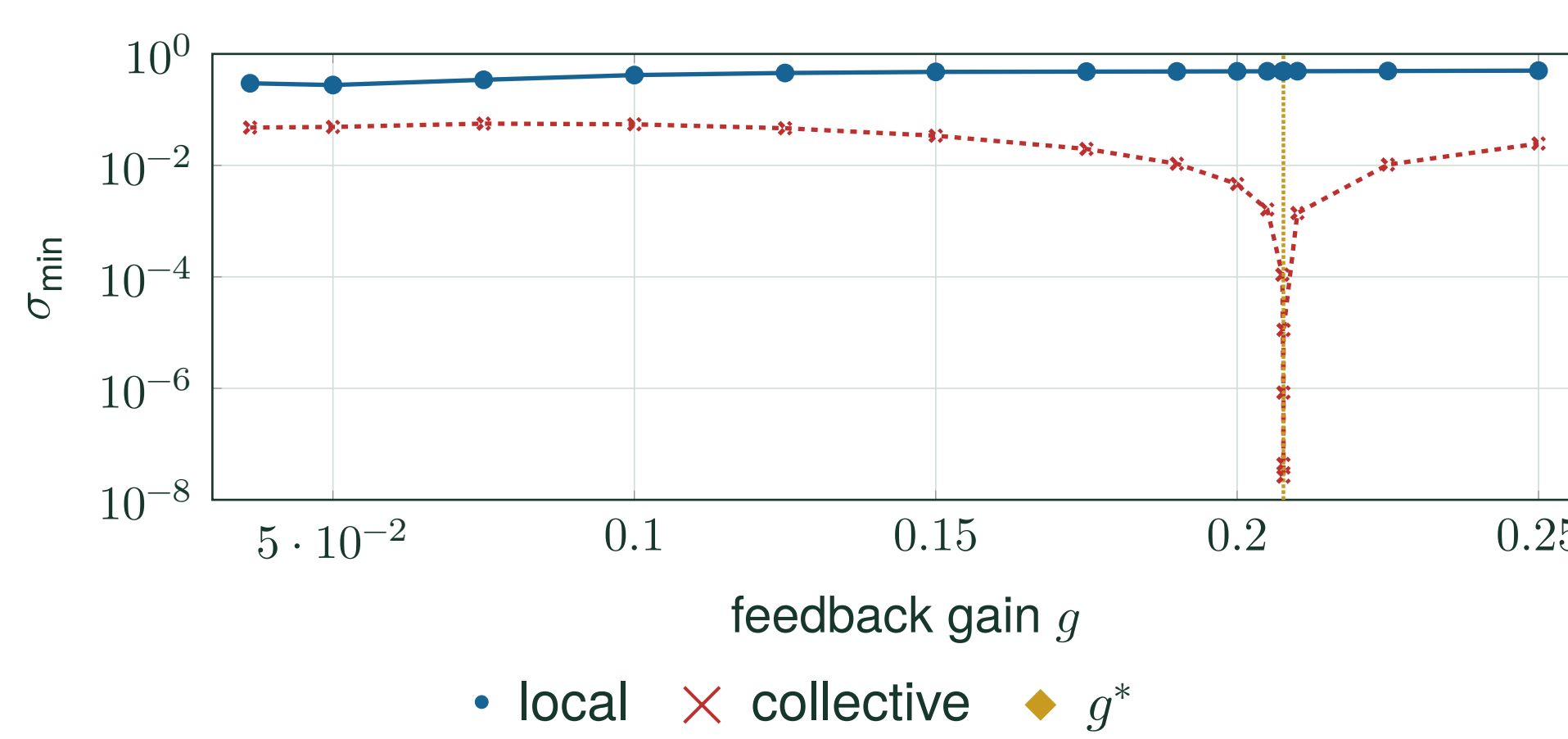
4A FROM DEVICES TO NETWORK RETURN

Network update	$T_S(s) = T_0(s) + \sum_{i \in S} E_i \Delta Y_i(s) E_i^T$
Characteristic	$\det P_S(s) = h_S(s) \det T_0(s) \det(I + M_{SS}(s))$
Local $\times$ collective	$\det(I + M_{SS}) = \left[\prod_{i \in S} \det(I + M_{ii}) \right] \det(I + Q_{SS})$
Interactions	$\det(I + Q_{SS}) = \sum_{T \subseteq S} \mu_T$

Strict M1 reduction is not established: residual 2.34×10^{-11} exceeds the 1.00×10^{-12} gate.

4B THE COLLECTIVE TERM CLOSES

Boundary scan: $g^* = 0.207681$; $\alpha_{\perp} = 0$; $f = 0.7064 \text{ Hz}$



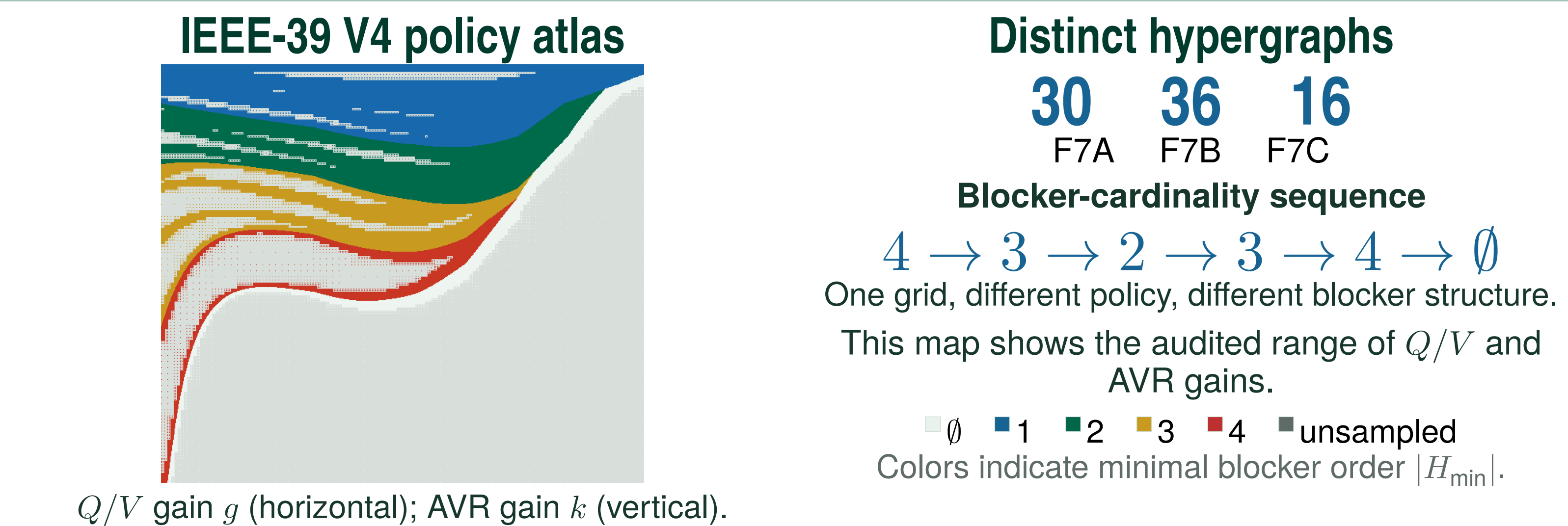
LOCAL DEVICES

$\sigma_{\min} \approx 0.49$
remain regular

NETWORK RETURN

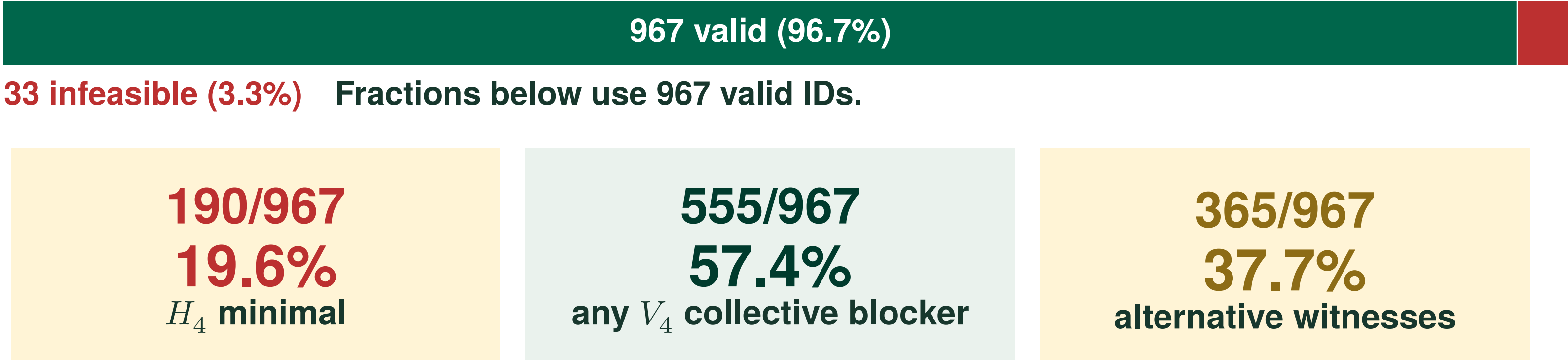
$\sigma_{\min} \approx 4.4 \times 10^{-8}$
approaches singularity

5 POLICY CHANGES THE MINIMAL BLOCKER



6 THE CANONICAL H_4 IS NOT UNIVERSAL

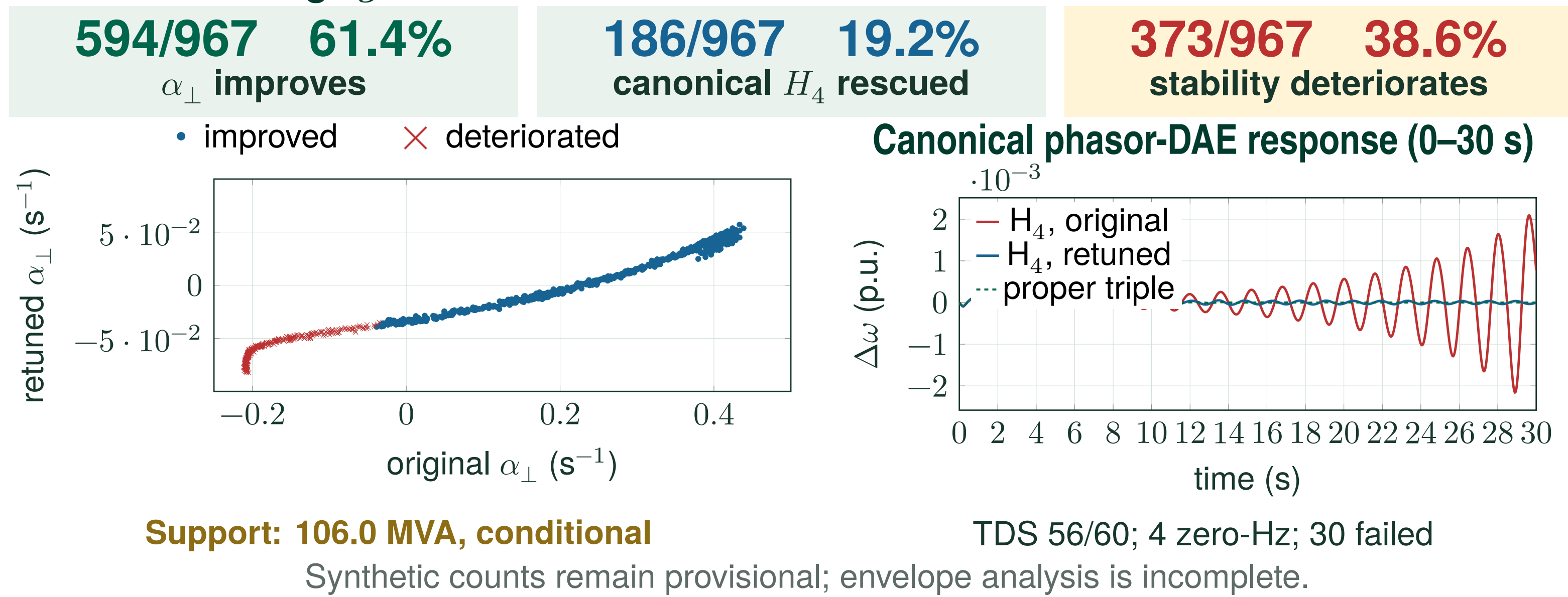
Frozen synthetic envelope (1000 scenario IDs)



Synthetic scenario totals are provisional pending review.
Descriptive synthetic ensemble; IDs are not a probability sample. H_4 is minimal in the audited event.

7 A CANONICAL REPAIR IS NOT UNIVERSAL

Control retuning: $g = 0.03625 \rightarrow 0.25$

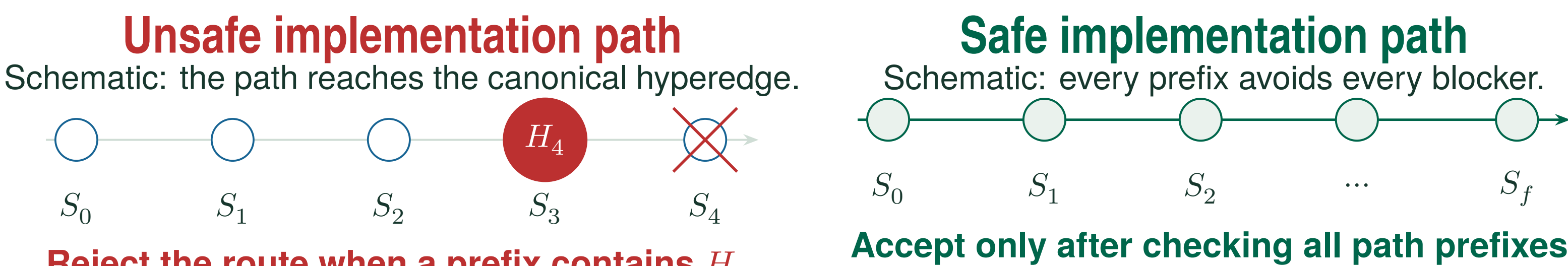


8 SAFE PATHS AVOID EVERY HYPEREDGE

Path-constrained planning formulation

$$\max_S \sum_{i \in S} P_i \quad \text{s.t. } e \not\subseteq S, \quad \forall e \in \mathcal{H}(\theta)$$

Apply the same constraint to every intermediate prefix of the implementation path.



The order matters: every prefix must avoid every blocker.

9A MODEL CHECKS

100/100
cross-code agreement
same model

56/60
TDS sign agreement
completed phasor-DAE runs

4 zero-Hz disagreements among 60 completed runs; 30 of 90 runs had solver failures.

9B WHAT “SAFE PATH” MEANS HERE

Each prefix is a **re-equilibrated phasor-DAE** portfolio.
The analysis does not simulate **physical switching**, EMT, current limits or DC-link limits.
Blockers depend on the model and control policy.

TAKEAWAY

SAFE PREFIXES, SAFE TRANSITIONS.

A stable endpoint can still hide an unstable intermediate coalition.

Blockers change with the control policy.

Safety here refers to re-equilibrated phasor-DAE prefixes.

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REFERENCES

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